Supplementary Material for “FAIR-BETH: A Fair Thresholding and Ablation Protocol for Behavioral Malicious-Process Detection under Distribution Shift”

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TABLE 1
Validation-derived thresholds.

Model	5%-FPR	Max-F1
IF-Platt	0.005286	0.005252
GRUAE-Platt	0.005408	0.005129
SimpleAvg	0.005318	0.005190
ScoreRF	0.059149	0.088814
TabRF	0.018254	0.174148
MetaRF	0.016541	0.195849

1 Feature Specification

For process i, endpoint events are grouped by `hostName` and `processId` and ordered by timestamp. The predictor excludes `sus` and `evil`. Scalar features comprise event count, unique event count, event entropy, argument-count mean, standard deviation and maximum, return-value mean and standard deviation, duration, event rate, unique argument strings, mean argument-string length, and unique parents. The fixed event histogram is fitted on the development pool only. For event identifier e,

$$h_{i,e} = \frac{1}{n_i}\sum_{j=1}^{n_i} \mathbf{1}\{eventId_{ij} = e\}.$$

$$H_i = -\sum_{e:h_{i,e}>0} h_{i,e}\log_2 h_{i,e}.$$

The event rate is

$$r_i = \frac{n_i}{(t_{i,n_i} - t_{i,1}) + 10^{-6}}.$$

Test-only event identifiers do not expand the fitted feature space.

2 Thresholds and Hyperparameters

Table 1 records the thresholds selected on `val_strat` for the principal ablation. Their small absolute values must not be interpreted as deployment probabilities.

The sequence capacity audit uses the first 512 tokens, batch size 128, Adam with learning rate 10^{-3}, at most eight epochs, and patience two. It predicts token $t+1$ from tokens through t; the target token is never fed to the decoder at the same position. The three tested configurations are GRU-64×1 (23 666 parameters), GRU-128×2 (183 218), and LSTM-128×2 (241 074).

TABLE 2
Principal model hyperparameters.

Component	Parameter	Value
Isolation Forest	Trees	200
Isolation Forest	Contamination	0.0050
GRU-AE	Embedding/hidden	32/64
GRU-AE	Layers/parameters	1/42 482
GRU-AE	Batch size/epochs	512/30
GRU-AE	Learning rate	10^{-3}
ScoreRF	Trees/minimum leaf	200/2
TabRF	Trees/input dimensions	200/61
MetaRF	Trees/input dimensions	300/64
RF-500	Trees/class weights	500/balanced
Bootstrap	Replicates	1 000

3 Repeated Threshold Procedure

The enriched development pool combines the training, calibration, and stratified-validation partitions and contains 5 819 processes, including 29 positives. Repeated stratified five-fold validation with ten repetitions produces 50 held-out score vectors. Each fold contains five or six positives. An RF-500 model is fitted independently in each fold and the largest-recall threshold satisfying empirical validation FPR ≤ 0.05 is stored. The median threshold is computed before any official-test evaluation. A final RF-500 is then fitted on the full development pool and evaluated once with the locked median threshold. Test-label access is not part of model fitting, threshold selection, or aggregation. Table 3 reports the resulting sensitivity across all preselected thresholds.

4 Reproducibility Checklist

The executable order is: read the public BETH CSV files; exclude DNS files; aggregate by host and process; derive

TABLE 3
Sensitivity over all 50 preselected thresholds on the fixed test score vector.

Metric	Median	Minimum	Maximum
Precision	0.7573	0.7411	0.7979
Recall	0.6446	0.6198	0.6860
F1	0.6964	0.6964	0.7124
Observed FPR	0.3247	0.2468	0.3766
False positives	25	19	29
False negatives	43	38	46

labels; remove label fields; fit the event vocabulary on development data; construct scalar, histogram, and sequence representations; create train, calibration, and validation partitions; train detectors and baselines; fit score calibration on calibration data; lock thresholds on validation data; evaluate the official test once; and write CSV, JSON, score, and figure artifacts. The repeated-threshold and sequence-capacity scripts are independent audits and do not overwrite the principal artifacts.

Authoritative generated files include `cross_fitted_thresholds.csv`, `locked_threshold_test_sensitivity.csv`, `sequence_capacity_ablation.csv`, and their JSON summaries. The complete package is available at https://github.com/EkodeckStephane/fair-beth-malicious-process-detection.

5 Data and Model Cards

Data purpose: offline evaluation of behavioral malicious-process detection under distribution shift. BETH is not a complete representation of production ransomware and lacks family labels. Supplementary BETH files are development data only; the official test remains the final BETH test. MalBehavD-V1 validates protocol portability to another schema, not direct transfer of a BETH-trained model.

Model purpose: the direct tabular forests are research benchmarks, not endpoint products. They use observable process statistics and event histograms. Their main limitation is threshold non-transfer: a 5% development FPR becomes 32.47% for the repeated-validation locked threshold and 36.36% for TabRF's principal threshold on the official test.

Failure disclosure: anomaly-score ranking inverts; score fusion does not beat the direct tabular baseline; recurrent next-event variants remain weak; early BETH prefixes provide low recall; calibration scores have no reliable posterior interpretation under test shift; host-disjoint threshold selection is unstable; and the official BETH test contains only 198 processes.

6 Supplementary Audit Figures

Figure 1 provides the calibration curves, Figures 2 and 3 report held-out permutation attribution, and Figure 4 reports the BETH prefix curve.

7 Deployment and Response Boundary

A live implementation must maintain rolling state per process and recompute features over prefixes. The paper's full-trace result cannot establish containment latency. Thresholds require target-environment calibration, host-role stratification, drift monitoring, and reversible response policies. Suggested actions range from evidence collection to network constraint; process termination requires corroborating signals and rollback safeguards. MITRE ATT&CK may organize analyst response, but BETH event identifiers are not used as technique labels because the dataset does not supply a validated mapping.

8 Extension Requirements

Direct transfer requires a schema-compatible endpoint dataset with benign and malicious processes, host and time metadata, and enough positive processes for independent calibration and testing. Family-specific claims additionally require verified family labels. A complete deployment study must report wall-clock detection delay, replay-based evasion, action reversibility, analyst workload, and predeclared cost functions. Oracle test thresholds may be shown only as sensitivity analyses and must never replace the locked operating point.

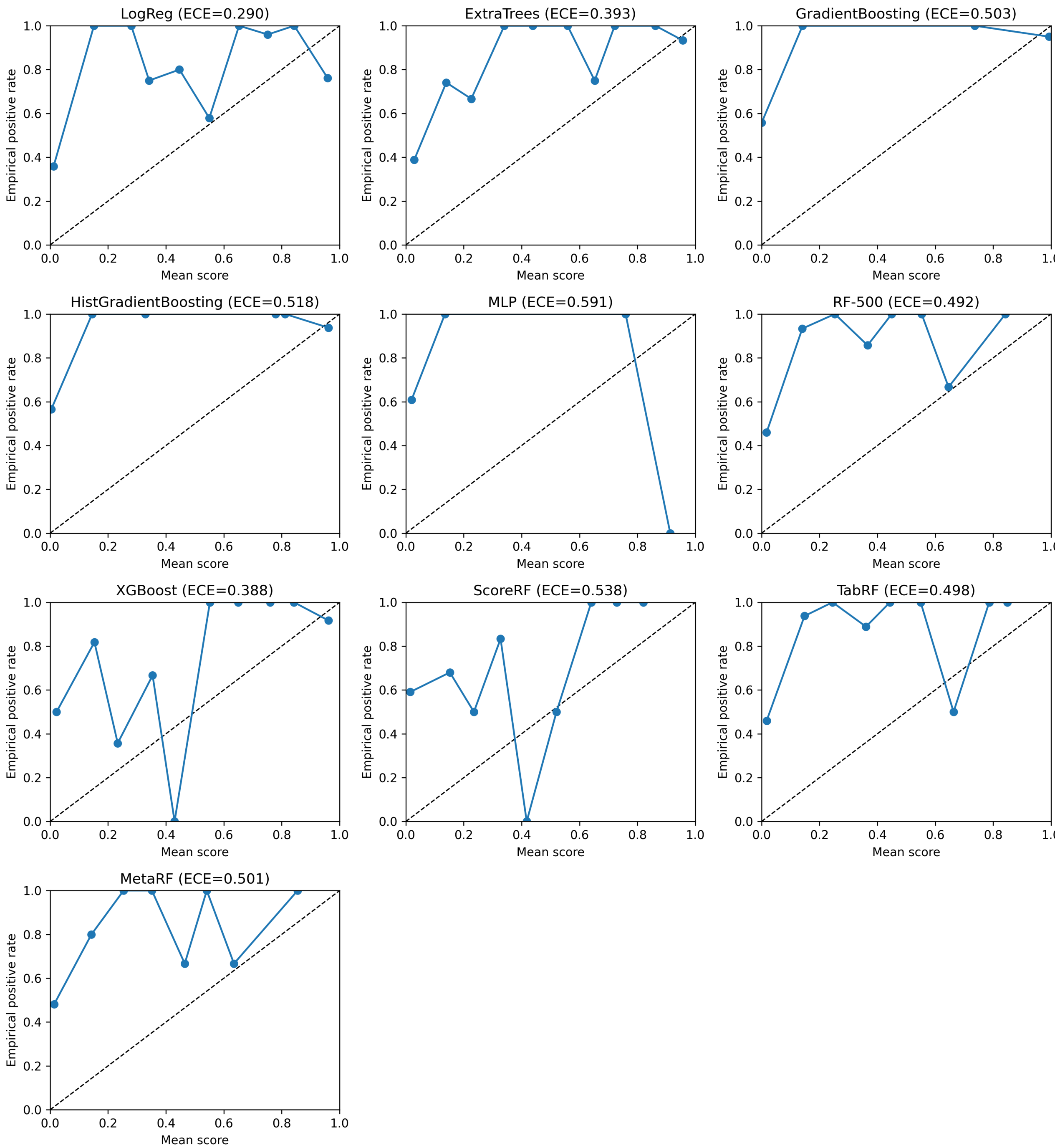

Fig. 1. Reliability diagrams on the official BETH test split. These curves support score-and-threshold language rather than posterior-risk language.

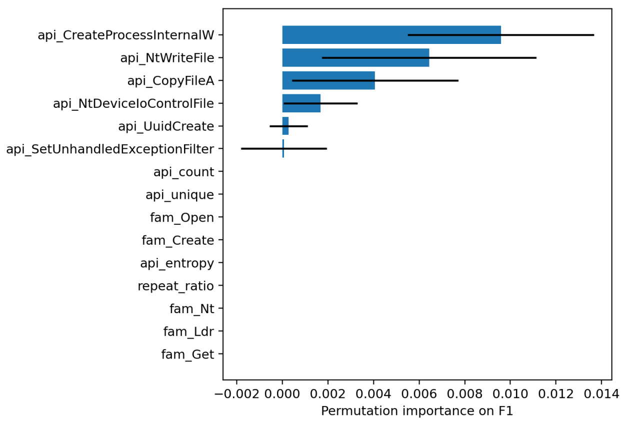

Fig. 2. Held-out permutation importance on MalBehavD-V1.

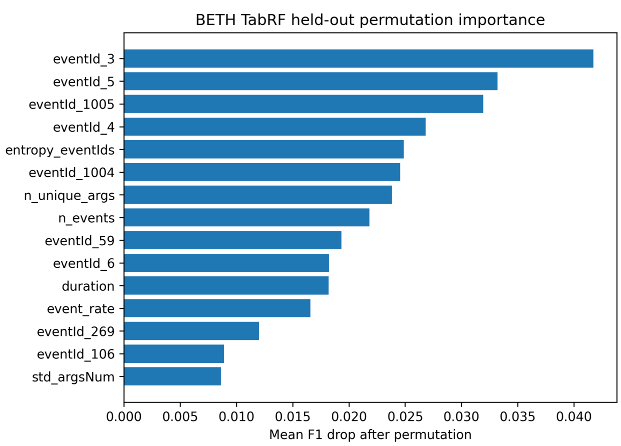

Fig. 3. Held-out permutation importance for BETH TabRF.

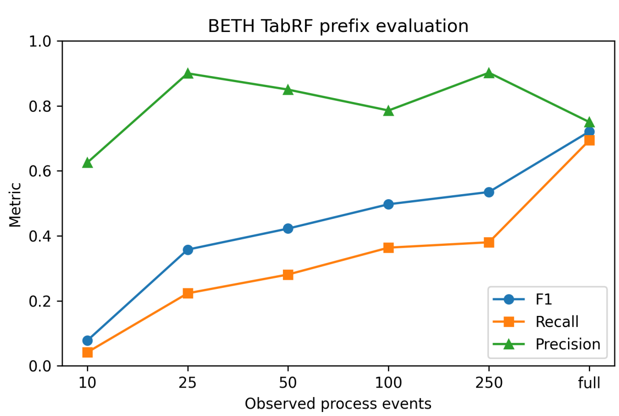

Fig. 4. BETH TabRF prefix evaluation.